

To'la differensialli tenglamalar. Agar

$$M(x, y)dx + N(x, y)dy = 0 \quad (1)$$

tenglamaning chap tomonidagi differensial ifoda potensial deb ataluvchi biror $u = u(x, y)$ funksiyaning to'la differensialidan iborat, ya'ni

$$du(x, y) = M(x, y)dx + N(x, y)dy \quad (2)$$

bo'lsa, u holda (1) tenglama **to'la (to'liq) differensialli tenglama** deyiladi. Bu yerdagi M va N funksiylar tekislikdagi biror D sohada aniqlangan. Biz

ularni hamda $\frac{\partial M}{\partial y}$ va $\frac{\partial N}{\partial x}$ xususiy hosilalarni uzluksiz deb hisoblaymiz.

Integrallovchi ko'paytuvchi. Aytaylik, (1) tenglama uchun bir bog'lamli D sohada (6) shart bajarilmasin. U holda bu tenglama to'la differensialli bo'lmaydi. Lekin biz uni D sohada nolga aylanmaydigan biror $\mu = \mu(x, y)$ silliq funksiyaga ko'paytirib, to'la differensialli tenglamani hosil qilishga harakat qilishimiz mumkin. Ushbu

$$\mu M dx + \mu N dy = 0 \quad (14)$$

tenglamani to'la differensialli bo'lishi uchun D sohada

$$\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N) \quad (15)$$

shart bajarilishi kerak. Bu yerdagi μ funksiya (1) tenglama uchun integrallovchi ko'paytuvchi deb ataladi. Oxirgi shartni

$$N \frac{\partial \mu}{\partial x} - M \frac{\partial \mu}{\partial y} = \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \mu \quad (16)$$

ko'rinishga keltiraylik. Bu tenglik μ funksiyaga nisbatan xususiy hosilali tenglamani ifodalaydi. (16) tenglamaning yechimlarini ba'zi hollarda osongina topish mumkin.

Faraz qilaylik, μ ni biror ma'lum $\omega(x, y)$ funksiyaning funksiyasi sifatida topish mumkin bo'lsin, ya'ni $\mu = \mu(\omega)$, $\omega = \omega(x, y)$, bo'lsin. Bu holda (16) tenglama

$$\frac{d\mu}{d\omega} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N \frac{\partial \omega}{\partial x} - M \frac{\partial \omega}{\partial y}} \cdot \mu$$

ko'rinishga keladi. Bu tenglama $\mu = \mu(\omega)$ ko'rinishdagi yechimga ega bo'lishi uchun

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$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N \frac{\partial \omega}{\partial x} - M \frac{\partial \omega}{\partial y}} = \psi \quad (17)$$

ifoda ω ning funksiyasi kabi ifodalanishi, ya'ni $\psi = \psi(\omega)$ bo'lishi kerak. Bu holda integrallovchi ko'paytuvchi uchun tenglama

$$\frac{d\mu}{d\omega} = \psi(\omega) \mu$$

ko'rinishni olib, u

$$\mu = e^{\int \psi(\omega) d\omega} \quad (18)$$

yechimga ega.

Shunday qilib, agar (17) ifoda faqat ω ning funksiyasi sifatida ifodalansa, u holda (1) tenglama uchun (18) ko'rinishdagi integrallovchi ko'paytuvchi mavjud.

Masalan, $\mu = \mu(x)$ ($\mu = \mu(y)$) ko'rinishdagi integrallovchi ko'paytuvchi mavjud bo'lishi uchun

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \psi \quad \left(\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} = \psi \right)$$

ifoda x o'zgaruvchiga (y o'zgaruvchiga) bog'liq bo'lmasligi, ya'ni u faqat x ga (y ga) bog'liq bo'lishi kerak. Bu holda

$$\mu(x) = e^{\int \psi(x) dx} \quad (\mu(y) = e^{\int \psi(y) dy})$$

ko'rinishdagi integrallovchi ko'paytuvchi mavjud.

Misol 2. Ushbu

$$(x^3 + y^2)dx + ydy = 0 \quad (20)$$

differentensial tenglamani yeching.

→ Berilgan tenglamada $M = x^3 + y^2$, $N = y$, $D = \mathbb{R}^2$. Demak,

$$\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2y.$$

Ushbu

$$\psi = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = 2$$

ifoda y ga bog'liq emas. Demak, berilgan tenglama $\mu = \mu(x)$ ko'rinishdagi integrallovchi ko'paytuvchiga ega. Bu $\mu = \mu(x)$ funksiya

$$\frac{d\mu}{dx} = \psi(x)\mu, \text{ ya'ni } \frac{d\mu}{dx} = 2\mu$$

tenglamani qanoatlantiradi. Oxirgi tenglamadan

$$\mu = e^{2x}$$

integrallovchi ko'paytuvchini topamiz. Endi berilgan tenglamani $\mu = e^{2x}$ ga ko'paytirib, to'la differensialli tenglama hosil qilamiz:

$$e^{2x}(x^3 + y^2)dx + e^{2x}ydy = 0.$$

Bu tenglamaning yechimi

$$u(x, y) = c$$

ko'rinishga ega. Bunda u potensial uchun

$$\frac{\partial u}{\partial x} = e^{2x}(x^3 + y^2), \quad (21)$$

$$\frac{\partial u}{\partial y} = e^{2x}y \quad (22)$$

bo'lishi kerak. (22) tenglikni y bo'yicha integrallab topamiz:

$$u = \frac{1}{2}e^{2x}y^2 + \varphi(x), \quad (23)$$

bunda $\varphi(x)$ hozircha noma'lum funksiya. (23) ni (21) ga qo'yib, $\varphi(x)$ uchun tenglama hosil qilamiz:

$$e^{2x}y^2 + \varphi'(x) = e^{2x}(x^3 + y^2) \text{ yoki } \varphi'(x) = e^{2x}x^3.$$

Bu munosabatdan bo'laklab integrallash yordamida noma'lum funksiya $\varphi(x)$ ni topamiz:

$$\begin{aligned} \varphi(x) &= \int e^{2x}x^3 dx = \frac{1}{2} \int x^3 d(e^{2x}) = \frac{1}{2}e^{2x}x^3 - \frac{1}{2} \int e^{2x}d(x^3) = \frac{1}{2}e^{2x}x^3 - \frac{3}{2} \int x^2 e^{2x} dx = \\ &= \dots = \frac{1}{2}e^{2x}x^3 - \frac{3}{4}e^{2x}x^2 + \frac{3}{4}e^{2x}x - \frac{3}{8}e^{2x}. \end{aligned}$$

Buni (23) ga qo'yib, u potensialni aniqlaymiz

$$u = \frac{1}{2}e^{2x}y^2 + \frac{1}{2}e^{2x}x^3 - \frac{3}{4}e^{2x}x^2 + \frac{3}{4}e^{2x}x - \frac{3}{8}e^{2x}.$$

Demak, berilgan (20) tenglamaning yechimlari

$$e^{2x}(4y^2 + 4x^3 - 6x^2 + 6x - 3) = c$$

tenglik bilan oshkormas ko'rinishda ifodalanadi. ♣

Ba'zan berilgan differensial tenglama uchun integral ko'paytuvchini tenglamani qismlarga ajratib, har bir qism uchun integrallovchi ko'paytuvchini topish yordamida qurish mumkin. Aytaylik, berilgan tenglamani ushbu

$$(M_1 dx + N_1 dy) + (M_2 dx + N_2 dy) = 0 \quad (19)$$

har bir qismining integrallovchi ko'paytuvchisi osongina topiladigan ko'rinishda yozish mumkin bo'lsin:

$$\mu_1 \cdot (M_1 dx + N_1 dy) = du_1,$$

$$\mu_2 \cdot (M_2 dx + N_2 dy) = du_2.$$

Ravshanki, μ_1 bilan birgalikda $\mu_1 \varphi(u_1)$ ham mos tenglamaning integrallovchi ko'paytuvchisidir, chunki

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$$\mu_1 \varphi(u_1) \cdot (M_1 dx + N_1 dy) = \varphi(u_1) du_1 = d(\varphi(u_1) du_1).$$

Buni hisobga olgan holda shunday $\varphi(u_1)$ va $\psi(u_2)$ funksiyalarni topaylikki, ular uchun

$$\mu_1 \varphi(u_1) = \mu_2 \psi(u_2)$$

munosabat o'rinli bo'lsin. U holda berilgan (19) tenglamaning har ikkala tomonini

$$\mu = \mu_1 \varphi(u_1) = \mu_2 \psi(u_2)$$

integrallovchi ko'paytuvchiga ko'paytirib,

$$\varphi(u_1) du_1 + \psi(u_2) du_2 = 0$$

tenglamani hosil qilamiz. Bu to'la differensialli tenglama osongina yechiladi va berilgan (19) tenglamaning yechimlari topiladi:

$$\int \varphi(u_1) du_1 + \int \psi(u_2) du_2 = c.$$

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